

Biruk Ambaye

Uppsala, Sweden | akibiruk39@gmail.com | biruk-embedded.vercel.app | github.com/Biruk-aki | +46 76 450 2691

Summary

Embedded Systems Master's student at Uppsala University with a strong foundation in low-level firmware engineering, real-time operating systems, and hardware-software integration. Experienced in developing RTOS-based applications in C/C++ across 32-bit microcontroller platforms, implementing standard communication protocols (I2C, SPI, UART, BLE), and building automated Python test pipelines for data acquisition and validation.

Education

Uppsala University

Master of Science in Embedded Systems

2025 – Present

Uppsala, Sweden

Mekelle University

Bachelor of Science in Computer Science

2017 – 2024

Mekelle, Ethiopia

Technical Skills

Languages: C, C++, Python, JavaScript (React.js)

Embedded & RTOS: Zephyr RTOS, FreeRTOS, ESP32-S3, ARM (Cortex-M33 / RP2350), Raspberry Pi Pico (RP2040), AVR (ATmega)

Protocols & Interfacing: I2C, SPI, UART, PWM, ADC, CAN bus, USB Serial

RF & Wireless: BLE (HOGP / GATT), Dyadic backscatter, 2-FSK modulation, sub-GHz (NRF52840, CC1352, CC2500), BER/PER analysis

Tools & Frameworks: PySerial, pytest, Logic Analyzers, Oscilloscopes, GDB, Git, Linux/POSIX, CMake, West

Embedded Projects

BLE Multi-Mode Desk Controller (ESP32-S3 HOGP Macro Knob)

2026

Personal Project | Zephyr RTOS, ESP32-S3, BLE HOGP, ADC, Devicetree, Kconfig

- Developed a driverless BLE desktop controller using **Zephyr RTOS** on an **ESP32-S3**, interfacing natively with host operating systems via **HID over GATT Profile (HOGP)**.
- Engineered a composite GATT server combining Consumer Control (volume, media, brightness) and Mouse Wheel descriptors over a single link with BLE Security Level 2 bonding.
- Implemented an ADC signal conditioning pipeline using 12 dB attenuation, an **Exponential Moving Average (EMA)** filter, and dynamic deadband thresholding to eliminate analog potentiometer noise.
- Structured a non-blocking architecture using Devicetree overlays, a 3-mode Finite State Machine, and asynchronous work-queue LED feedback.

Dyadic Backscatter Communication – Baud Rate Optimization

2026

Research Project | Uppsala University | NRF52840, Raspberry Pi Pico, CC1352, CC2500

- Built an experimental backscatter testbed with NRF52840 as carrier, Raspberry Pi Pico with dual CC2500 antennas as passive 2-FSK tag, and CC1352 as receiver.
- Evaluated baud rate reduction (100 kBaud → 50 kBaud) to improve SNR and link reliability without raising transmit power.
- Authored automated Python scripts to parse 15 measurement logs, evaluate Bit Error Rate (BER)/Packet Error Rate (PER), and validate hardware sync-word rejection.

Multi-Node Health Monitoring System

2026

Course Project | Zephyr RTOS, Raspberry Pi Pico 2, KY-039, MIKROE-3102 SpO2, BME680, ADC, I2C, UART

- Built a two-node monitoring setup running Zephyr RTOS to acquire heart rate (ADC), SpO2 (I2C), and ambient temperature (I2C).
- Implemented inter-node UART streaming to a base station with real-time serial visualization on a host PC and automated regression tests via pytest.

Environmental Monitoring & Hardware Alert System

2026

Course Project | Zephyr RTOS, RP2350 (ARM Cortex-M33)

- Developed BME680 sensor firmware managing MOX heater plate cycling to evaluate Gas Resistance and IAQ scores.
- Integrated Devicetree overlays (.overlay) and Kconfig configurations for sensor APIs and software PWM alert drivers.

Deterministic Real-Time Scheduler
Academic Project | Embedded C, AVR (ATmega)

2025

- Developed a priority-based deterministic scheduler for AVR microcontrollers to manage periodic real-time tasks.
- Performed Worst-Case Execution Time (WCET) analysis to guarantee hard real-time deadline compliance.

Experience

Aynalem Electronics
Junior Frontend Web Developer

2024 – 2025
Mekelle, Ethiopia

- Developed modular React.js interfaces adhering to structured code review and quality standards.
- Optimized front-end performance and state management logic across web platforms.

Proxima Technologies
Frontend Web Developer

2023 – 2024
Mekelle, Ethiopia

- Collaborated in agile Scrum sprints utilizing Git version control for continuous integration and documentation.

Languages

English: Professional working proficiency | **Swedish:** Basic (Actively Studying) | **Amharic / Tigrigna:** Native